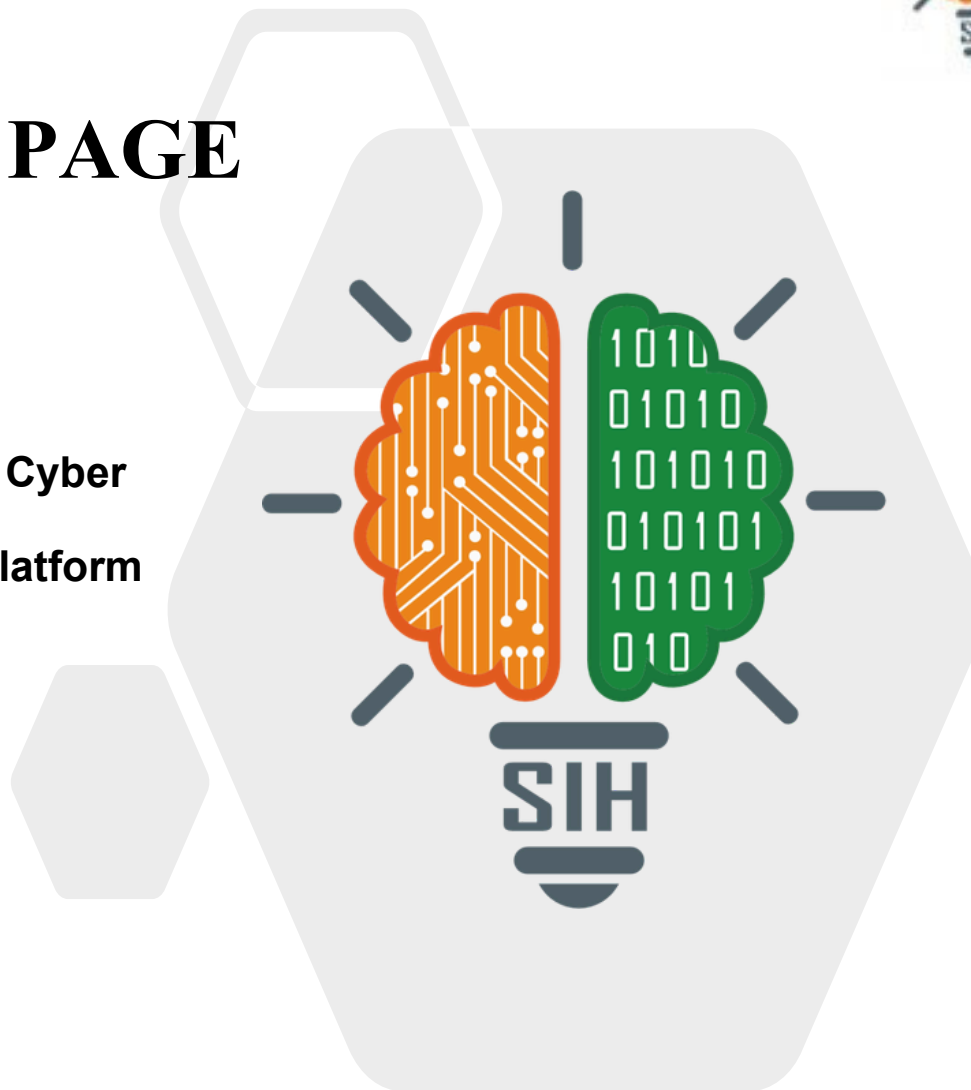


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – 26105
- Problem Statement Title – AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform
- Theme – Blockchain & Cybersecurity
- PS Category- Software
- Team ID –
- Team Name (Registered on portal) – a2z



Proposed Solution

- **Verified Exposure Graph** - Confidence-tagged attack graph (Verified / Configured / Estimated) replaces a black-box risk score.
- **Choke-Point Optimizer** - Set-cover/LP engine ranks fixes by real risk removed per rupee — no double-counted spend.
- **Exploit-Aware Likelihood** - Risk recalculates instantly on live CISA KEV / EPSS signals, not on a quarterly cycle.
- **Living Regulatory Twin** - Auto-maps every control to ISO 27001, NIST CSF, CIS, RBI 2026 & SEBI CSCRf, with equivalence logic.

Unique Value Proposition

- **Evidence-Graded Trust** - Every ₹ figure traces back to real telemetry, not a self-reported guess.
- **Mathematically Bounded ROI** - Spend recommendations capped by the Gordon-Loeb ceiling — never overspend the risk.
- **India-Native Compliance** - Multi-regulator equivalence, not a NIST checklist with an RBI tab bolted on.
- **Continuous, Not Annual** - The number moves the same day a vulnerability turns exploitable.

From Security Noise To A Board Decision

01

Ingest Signals, assets, CVE scans and telemetry

02

Verify Trace paths and tag evidence confidence

03

Quantify Risk quantification with FAIR + Monte Carlo

04

Optimize Compare control cost, reduction and return

05

Govern Turn the answer into board-ready action

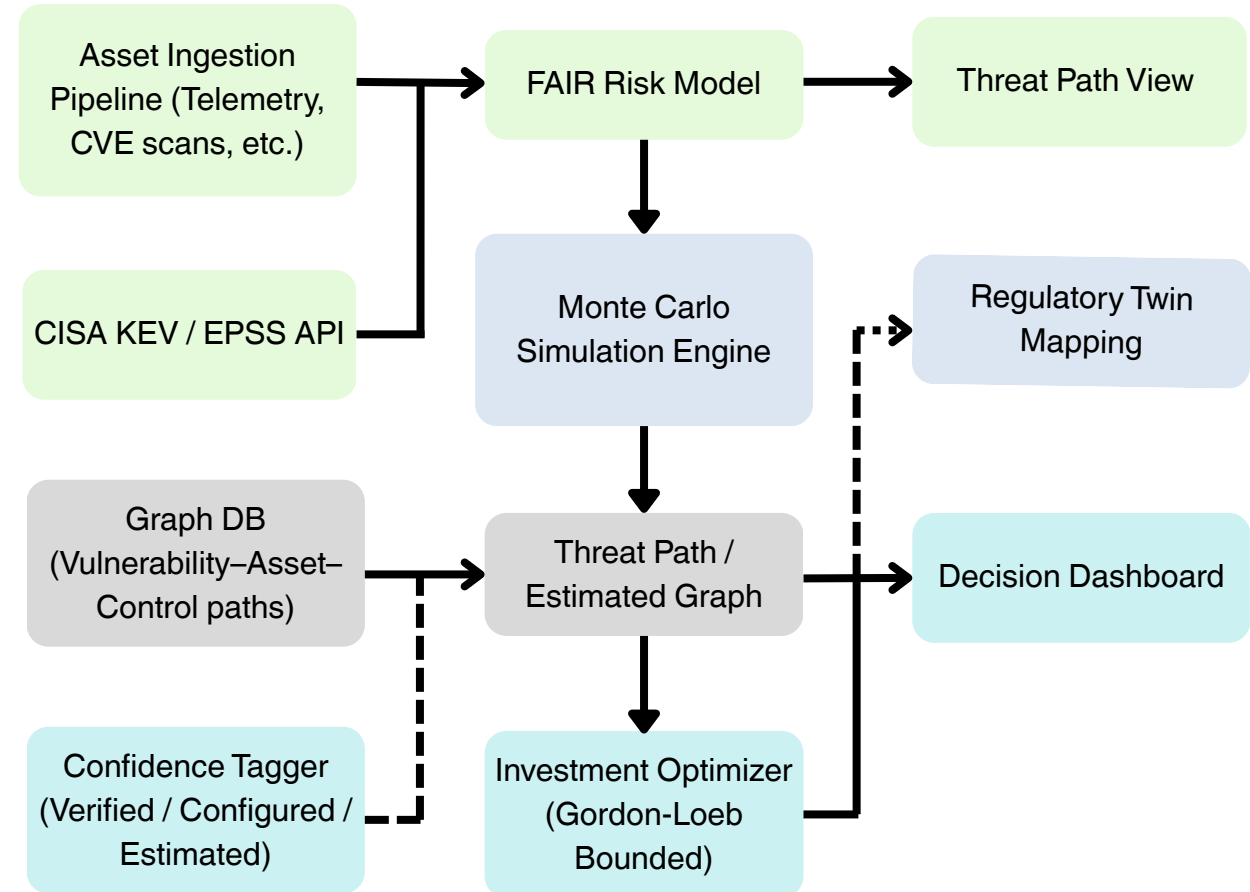
TECHNICAL APPROACH

Technology Stack

1. Frontend - Next.js + React
2. Backend - FastAPI + Python
3. Persistence - PostgreSQL via SQLAlchemy/asyncpg
4. Caching - Redis
5. Quantification - Python Monte Carlo engine with FAIR-inspired risk aggregation
6. Optimization - Custom Gordon-Loeb-bounded optimizer
7. Threat intelligence - KEV/EPSS data model and integration-ready fields



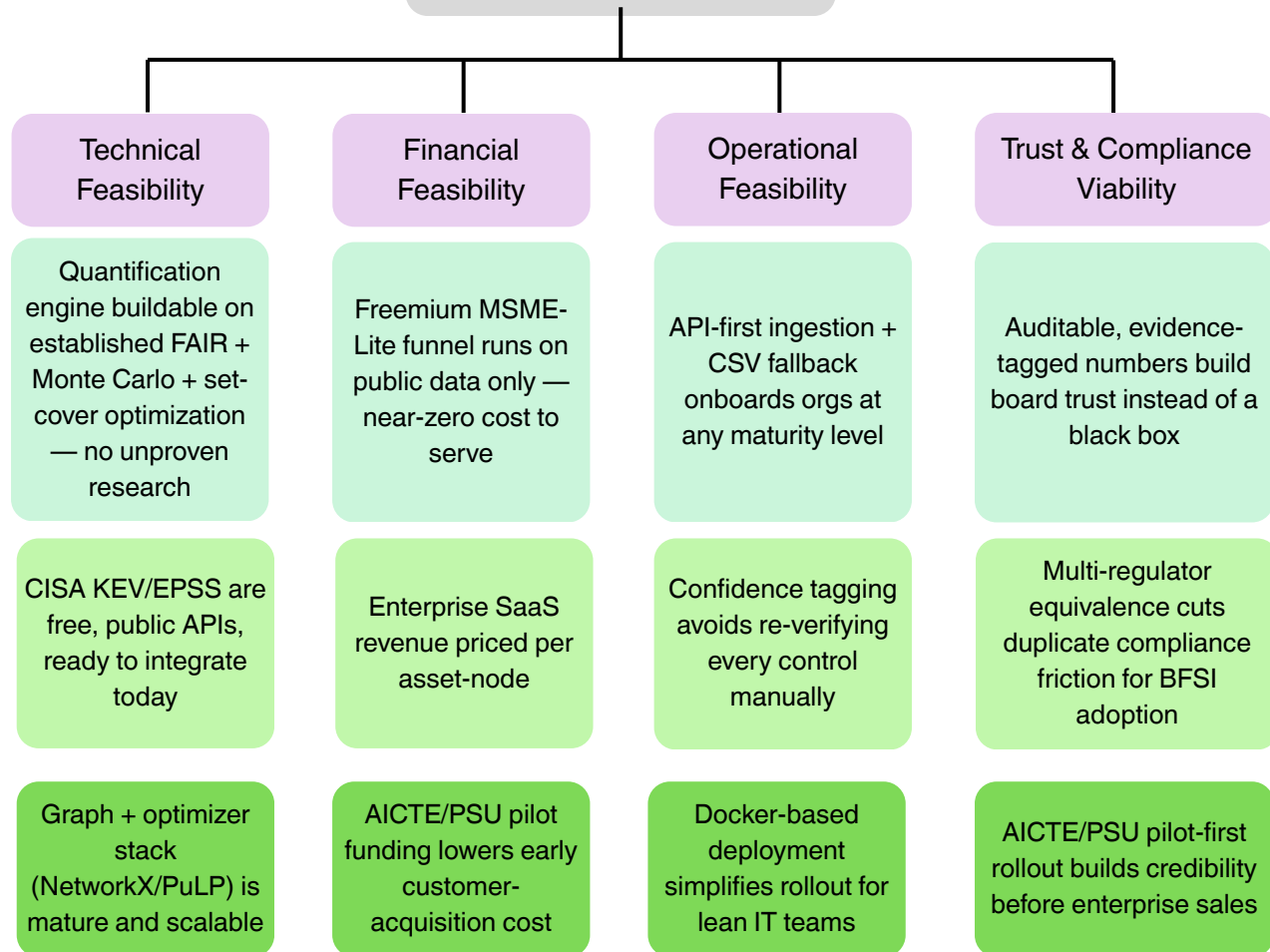
Architecture Diagram



FEASIBILITY AND VIABILITY



Feasibility Matrix



Strategies for Overcoming Challenges

Technical Challenges

- Data quality varies early on → mitigated by honest confidence tagging, not a hidden average
- Optimizer is NP-hard at scale → mitigated via greedy/LP-relaxation, no false optimality claims
- Live feed outages → mitigated by caching last-known-good KEV/EPSS state

Financial Challenges

- Long BFSI sales cycles → offset by free MSME-Lite funnel + PSU pilots for early traction
- Thin Indian breach-cost data → mitigated via hierarchical Bayesian priors blending global and local evidence

Trust & Compliance Challenges

- New-entrant skepticism → mitigated by publishing an inspectable methodology, not just a claim
- Regulatory drift (RBI/SEBI updates) → mitigated by human-reviewed change-diffs, never autonomous remapping

IMPACT AND BENEFITS



Benefits and Solution

**01**

Reduced Audit Overlap

SEBI/RBI multi-regulator equivalence logic eliminates redundant control testing.

**02**

Explainable Risk

Confidence tags and node-level evidence make every number inspectable, not asserted.

**03**

Non-Overlapping Prioritization

Choke-point ranking finds the one fix that closes multiple attack paths at once.

**04**

Actionable Threat Intelligence

Live likelihood recompute from EPSS/KEV signals enables same-day mitigation.

Business Model

MSME-Lite / Freemium
Public-data-only graph, wide funnel, entry-level adoption.

Enterprise BFSI SaaS
Full telemetry ingestion, verified confidence tiers, priced by asset-node.

Pilot Programs
AICTE/PSU pilots build government-linked trust as a first step.



Target Customers

- MSMEs and mid-market enterprises (via the freemium/MSME-Lite tier)
- Indian BFSI (commercial banks, NBFCs, payments banks)
- Government agencies and PSUs (via the RBI Directions crosswalk)



Impacts on the Target Audience

Risk Reduction

Prevents double spending; targets choke points.

Regulator Confidence

Converts telemetry into rupee impact.

Smart Security Spend

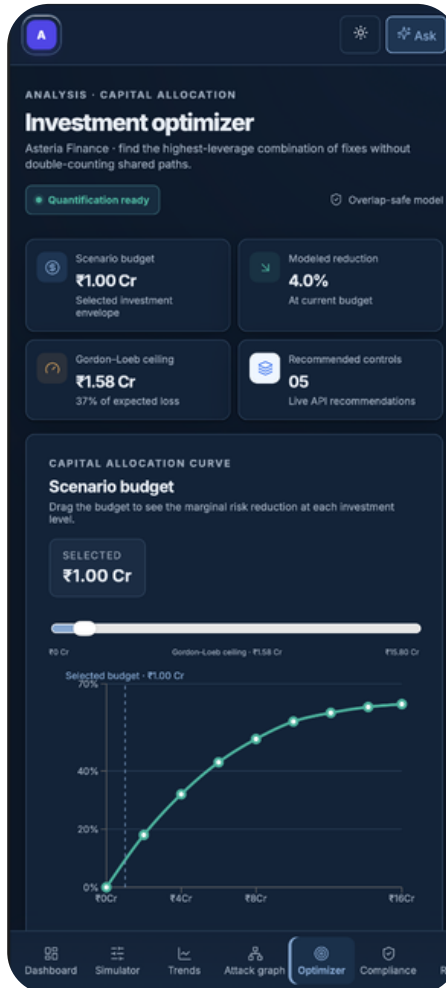
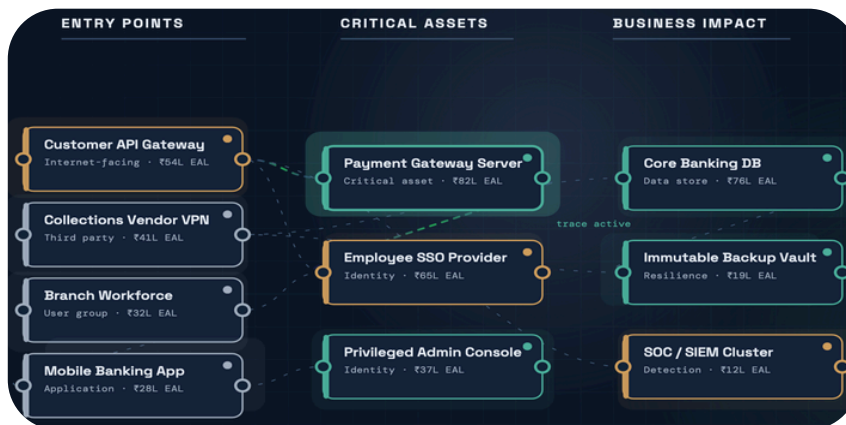
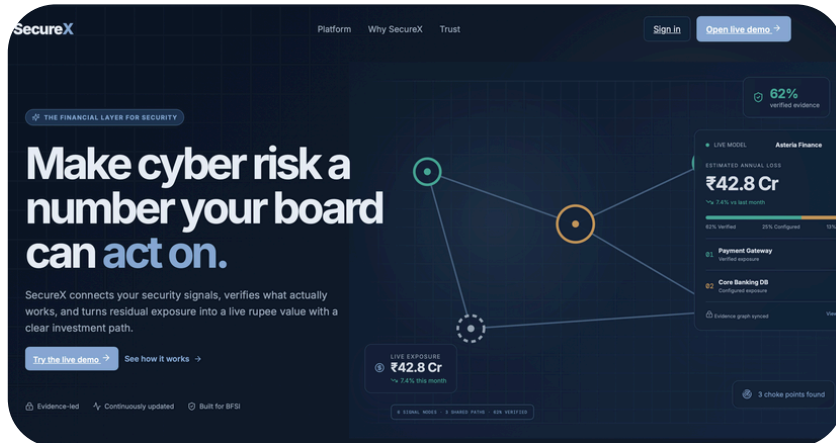
Enables defensible investments.

Live Compliance
Maps RBI/SEBI 2026 rules.

RESEARCH AND REFERENCES



PROTOTYPE UI



Research Foundation

- Open FAIR — Quantitative risk model based on loss event frequency and loss magnitude. [The Open Group — Open FAIR](#)
- CISA KEV — Catalog of vulnerabilities confirmed to be exploited in the wild, supporting risk-based prioritization. [CISA Known Exploited Vulnerabilities Catalog](#)
- EPSS — Daily probability score estimating the likelihood that a publicly disclosed vulnerability will be exploited. [FIRST EPSS](#)
- Monte Carlo Simulation — Represents uncertainty by analyzing many possible risk outcomes rather than relying on one point estimate. [NIST SP 800-30 Risk Assessment Guidance](#)
- Gordon-Loeb Model — Provides an economic basis for determining rational cybersecurity investment levels. [Gordon & Loeb, ACM Research Paper](#)
- NIST CSF 2.0 — Provides a governance and communication structure for managing cybersecurity risk. [NIST Cybersecurity Framework 2.0](#)

SecureX integrates these established methods into an evidence-aware, board-level investment decision.